AGE 310: Instrumentation and Measurements

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Abstract

Lecture 3 provides a detailed examination of resistive transducers, focusing on the fundamental principle that resistance changes with physical stimuli. We analyze Potentiometers for displacement, Strain Gauges for mechanical stress (force), and RTDs and Thermistors for temperature measurement, alongside the essential signal conditioning circuit: the Wheatstone Bridge.

Lecture 3: Resistive Transducers

3.1 Fundamental Principle of Resistive Sensing

The operation of all resistive transducers is based on the dependency of a conductor's resistance (R) on its physical dimensions and material properties. The fundamental formula is:

$$R = \rho \frac{L}{A}$$

Where:

- R: Electrical Resistance (Ω)
- ρ: Resistivity ($\Omega \cdot$ m) - Material property (e.g., affected by temperature or light).
- L: Length of the conductor (m) - Affected by displacement or strain.
- A: Cross-sectional Area (m^2) - Affected by strain.

Resistive transducers work by changing one of these three factors: L, A, or ρ.

3.2 Potentiometers (Pots)

3.2.1 Principle of Operation

A potentiometer is a passive transducer that converts linear or angular displacement into a measurable voltage or resistance change. It operates on the principle of varying the length (L) of the resistive element by moving a contact (wiper).

The output voltage (V_{out}) is determined by the voltage divider rule:

$$V_{\text{out}} = V_{\text{in}} \left(\frac{R_{\text{wiper}}}{R_{\text{total}}} \right) = V_{\text{in}} \left(\frac{x}{x_{\text{max}}} \right)$$

Where x is the current position and x_{max} is the full range of travel.

3.2.2 Types and Applications

- **Linear Potentiometers:** Used to measure linear extension or travel.
  - **Ag Application:** Measuring the exact extension of a hydraulic cylinder on a tractor's front-end loader or linkage.
- **Rotary Potentiometers:** Used to measure angular rotation (e.g., 0° to 360°).
  - **Ag Application:** Measuring the steering angle of an autonomous vehicle or the rotational position of a boom sprayer section.

3.2.3 Loading Effect Error

Ideally, the output is perfectly linear. However, when the output of the potentiometer is connected to a measuring circuit (the Load, R_{load}), it draws current. This places R_{load} in parallel with a portion of the potentiometer's resistance (R_{wiper}), causing a non-linear voltage drop known as the **Loading Error**.

$$R_{\text{eff}} = \frac{R_{\text{wiper}} \cdot R_{\text{load}}}{R_{\text{wiper}} + R_{\text{load}}}$$

Mitigation: To ensure linearity, the input impedance of the measuring circuit (R_{load}) must be significantly higher ($\geq 100\times$) than the total resistance of the potentiometer (R_{total}). This is achieved using buffer amplifiers (Op-Amps) in the signal conditioning stage.

3.3 Strain Gauges

3.3.1 Principle of Operation

A strain gauge is a small, foil-based sensor used to measure **Strain** (ε) which is a measure of deformation ($\Delta L/L$). When attached to a machine component (like a tractor drawbar or load cell beam), the gauge stretches or compresses along with the component.

Strain is related to the change in resistance (ΔR) via the Gauge Factor (GF).

Geometric Effect: When stretched: L increases, A decreases. Both effects increase the resistance (R).

3.3.2 The Gauge Factor (GF)

The Gauge Factor is the sensitivity of the strain gauge material and construction.

$$GF = \frac{\text{Relative change in Resistance}}{\text{Strain}} = \frac{\Delta R/R}{\varepsilon}$$

- **Metallic Foil Gauges:** GF is typically low, around 2.0 to 2.2.
- **Semiconductor Gauges:** GF is very high (≈ 100), utilizing the *piezoresistive effect* (change in ρ due to stress), making them extremely sensitive but highly susceptible to temperature variation.

Calculation Example 1: Strain Gauge Output

A strain gauge with a nominal resistance of $R = 120\ \Omega$ and a $GF = 2.1$ is subjected to a strain of $\varepsilon = 500$ µstrain ($\mu\varepsilon$, microstrain, or 500×10^{-6}). Calculate the change in resistance (ΔR).

Solution: First, rearrange the GF formula: $\Delta R = GF \cdot R \cdot \varepsilon$.

$$\Delta R = 2.1 \times 120\ \Omega \times (500 \times 10^{-6})$$

$$\Delta R = 2.1 \times 0.06\ \Omega = 0.126\ \Omega$$

The change in resistance is only 126 mΩ. This tiny change necessitates a precise measurement circuit.

3.4 The Wheatstone Bridge

3.4.1 Necessity

Because the resistance change (ΔR) in a strain gauge is very small, we cannot use a standard ohmmeter. We need a circuit that converts this small change into a measurable voltage (ΔV). The Wheatstone Bridge is the standard solution.

3.4.2 Circuit Principle

The bridge consists of four resistors (R_1, R_2, R_3, R_4) connected in a diamond shape, excited by a voltage (V_{in}). The output voltage (V_{out}) is measured across the middle of the bridge.

The bridge is **balanced** when:

$$\frac{R_1}{R_2} = \frac{R_4}{R_3} \quad \Longrightarrow \quad V_{\text{out}} = 0$$

When one resistor (R_1, the Active Strain Gauge) changes resistance by ΔR, the bridge becomes unbalanced, and a proportional output voltage (V_{out}) appears. [Image of Wheatstone bridge circuit diagram showing the four resistors, input voltage, and output voltage]

3.4.3 Bridge Configurations and Compensation

1. **Quarter Bridge:** Uses 1 Active Gauge and 3 fixed resistors. Simple, but suffers from temperature drift, as temperature affects the gauge's resistance (R) directly.
2. **Half Bridge:** Uses 2 Gauges (one active, R_1, and one dummy, R_2, placed orthogonally or inactive). The temperature effect on R_1 is cancelled by the equal temperature effect on R_2. Output sensitivity is doubled.
3. **Full Bridge:** Uses 4 Gauges, all active and arranged to maximize the strain difference ($\pm\Delta R$).

Load Cells: Commercial load cells (used for weighing silos or livestock) universally employ a **Full Bridge** configuration for maximum sensitivity and inherent temperature compensation.

Calculation Example 2: Wheatstone Bridge Output

A quarter bridge strain gauge circuit is used with $V_{\text{in}} = 5$ V, $R = 120\ \Omega$, and $\Delta R = 0.126\ \Omega$ (from Example 1). The bridge output voltage (V_{out}) for a small ΔR is approximated by:

$$V_{\text{out}} \approx V_{\text{in}} \cdot \frac{1}{4} \cdot \frac{\Delta R}{R}$$

Solution:

$$V_{\text{out}} = 5\ \text{V} \cdot \frac{1}{4} \cdot \frac{0.126\ \Omega}{120\ \Omega}$$

$$V_{\text{out}} = 1.25\ \text{V} \cdot (0.00105) \approx 0.00131\ \text{V} = 1.31\ \text{mV}$$

The output is only 1.31 mV. This weak signal must be amplified by the signal conditioning stage (Lecture 6) before ADC.

3.5 Resistive Temperature Transducers

The resistance of many materials changes predictably with temperature (ρ changes).

3.5.1 Resistance Temperature Detectors (RTDs)

- **Material:** Pure metals (Platinum, Nickel, Copper). Platinum is preferred due to its stability and linearity.
- **Operation:** Resistance increases linearly with temperature (Positive Temperature Coefficient, PTC).
- **Standard:** The PT100 is the industry standard (Platinum, $R = 100\ \Omega$ at 0°C).
- **Pros:** High accuracy, excellent linearity, wide temperature range (e.g., -200°C to 850°C).
- **Cons:** Expensive, low sensitivity (small ΔR per degree), and often require complex signal conditioning (like the Wheatstone Bridge) due to low R.

3.5.2 Thermistors

- **Material:** Semiconductor metal oxides (ceramic).
- **NTC (Negative Temperature Coefficient):** Resistance **decreases** as temperature increases (most common type). Heat liberates electrons, increasing conductivity.
- **PTC (Positive Temperature Coefficient):** Resistance **increases** sharply at a critical temperature (often used as self-resetting fuses).
- **Pros:** Very high sensitivity (large ΔR per degree), cheap, and fast response time.
- **Cons:** Highly non-linear (exponential relationship), narrow operating range, requires linearization in the control system.

Calculation Example 3: RTD Linear Approximation

The resistance of a PT100 RTD is approximated linearly by $R_T = R_0(1 + \alpha T)$, where $R_0 = 100\ \Omega$ and α (temperature coefficient) $= 0.00385/^\circ$C. Calculate the resistance at 50°C.

Solution:

$$R_{50^\circ\text{C}} = 100\ \Omega \cdot (1 + 0.00385/^\circ\text{C} \cdot 50^\circ\text{C})$$

$$R_{50^\circ\text{C}} = 100\ \Omega \cdot (1 + 0.1925) = 119.25\ \Omega$$

3.6 Resistive Sensors in the Shamba (Farm)

Gypsum Blocks (Electrical Resistance Blocks):

- **Principle:** Two electrodes embedded in a porous block of gypsum. The block absorbs water from the soil until equilibrium.
- **Sensing:** As the block becomes wetter, its electrical resistance drops drastically.
- **Ag Use:** Measuring soil matric potential (soil water stress), particularly useful for scheduling irrigation in traditional settings.
- **Limitation:** The gypsum dissolves over time and must be replaced. Requires AC excitation to prevent electrode polarization.

3.7 Summary of Key Terms (Swahili/Context)

- **Potentiometer:** Kipimo cha uhamaji (Displacement measurement device).
- **Strain Gauge:** Kipimo cha mkazo (Measures mechanical stress/deformation).
- **Load Cell:** Kiini cha kipimo cha uzito (The device containing strain gauges, used for weighing).
- **Wheatstone Bridge:** Mzunguko wa usawazishaji (The circuit used to measure small resistance changes).
- **RTD:** Kichunguzi cha Halijoto kwa kutumia Upinzani (Resistance Temperature Detector).
- **NTC Thermistor:** Kichuna Halijoto cha NTC (Resistance decreases as temperature rises).
- **Gypsum Block:** Kizuizi cha Jasi (Resistive soil moisture sensor).

Lecture 3: Review Questions

SECTION A: Basic Concepts (5 Questions)

1. Explain the fundamental principle $R = \rho(L/A)$. Which variable does a potentiometer manipulate to achieve measurement?
2. Define the term *Gauge Factor* (GF) of a strain gauge. What is the typical range for metallic foil gauges?
3. State two distinct disadvantages of using a potentiometer for high-precision displacement measurements.
4. Differentiate between an RTD and an NTC Thermistor based on (a) material composition and (b) linearity of the output.
5. Explain why the Wheatstone Bridge circuit is essential for signal conditioning when using a strain gauge.

SECTION B: Advanced Analysis and Calculations (10 Questions)

6. **Strain Gauge Bridge Configuration:** Explain the principle of *temperature compensation* achieved by using a Half Bridge (2-gauge) configuration for strain measurement. Where must the second gauge (the dummy gauge) be placed, and what role does it play?
7. **Potentiometer Loading Error Calculation:** A 10 kΩ linear potentiometer is excited by a 10 V source. The output is measured by a voltmeter with an input impedance of 50 kΩ. Calculate the theoretical output voltage (V_{out}) at 50% travel, ignoring the loading effect, and then calculate the actual V_{out} considering the loading effect. State the percentage error caused by loading.
8. **Load Cell Force Measurement:** A load cell uses a $GF = 2.0$ full bridge configuration. When loaded with 1000 N, the strain is measured as 300 µstrain. If the bridge is excited with 5 V, estimate the output voltage (V_{out}). *Hint: A full bridge triples the quarter bridge sensitivity, but strain gauges are typically wired to produce $4\times$ output.* Assume the quarter bridge constant of $\frac{1}{4}$ applies relative to $\frac{\Delta R}{R}$.
9. **Thermistor Application:** An NTC thermistor is used to monitor engine coolant temperature. Explain the physical reason why its resistance drops non-linearly when temperature increases. If the resistance range is 10 kΩ at 20°C and 1 kΩ at 80°C, what is the sensitivity over this range?
10. **Gypsum Block Limitation:** Explain why an AC (alternating current) voltage, rather than a DC (direct current) voltage, must be used to excite a Gypsum Block soil moisture sensor. What physical process would DC excitation cause that corrupts the measurement?
11. **PTC Thermistor Application:** Explain a specific non-temperature measurement application for a PTC thermistor (Positive Temperature Coefficient) in a piece of agricultural machinery, and how its characteristic is exploited.
12. **RTD 3-Wire Compensation:** In highly precise RTD measurements, a 3-wire connection is often used instead of a 2-wire connection. Explain why the 3-wire system is preferred, mentioning the type of systematic error it cancels out.

13. **Drift in Strain Gauges:** Explain the difference between *Strain* and *Creep* (or *Drift*) in a strain gauge application. Which characteristic of the material (metal beam) or the gauge is responsible for creep?

14. **Linearization Necessity:** Given that the relationship for an RTD is nearly linear but a Thermistor is exponential, which sensor would require more computational power from a microcontroller (e.g., Arduino) to convert resistance readings accurately into temperature, and why?

15. **Zero Error Mitigation:** A linear potentiometer used in a boom sprayer reads 0.5 V when the boom is fully retracted (should be 0 V). This is a systematic error. Describe two practical methods (one electronic, one mechanical) to eliminate this 0.5 V zero error.